Research Paper

PAIN

Human induced pluripotent stem cell-derived GABAergic interneuron transplants attenuate neuropathic pain

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Abstract

Neuropathic pain causes severe suffering, and most patients are resistant to current therapies. A core element of neuropathic pain is the loss of inhibitory tone in the spinal cord. Previous studies have shown that foetal GABAergic neuron precursors can provide relief from pain. However, the source of these precursor cells and their multipotent status make them unsuitable for therapeutic use. Here, we extend these findings by showing, for the first time, that spinally transplanted, terminally differentiated human induced pluripotent stem cell-derived GABAergic (iGABAergic) neurons provide significant, long-term, and safe relief from neuropathic pain induced by peripheral nerve injury in mice. Furthermore, iGABAergic neuron transplants survive long term in the injured spinal cord and show evidence of synaptic integration. Together, this provides the proof in principle for the first viable GABAergic transplants to treat human neuropathic pain patients.

Keywords: Human iPSC, Neuropathic Pain, Peripheral Nerve Injury, GABAergic interneurons, Spinal transplantation, Stem cell therapy

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1. Introduction

Chronic pain has an increasing global impact, with pain representing a substantial burden of disease.[16] Chronic neuropathic pain is a serious subtype of pain caused by a lesion to the peripheral or central nervous system, typically leading to cold and tactile allodynia (the sensation of pain to normally innocuous stimuli).[8] Neuropathic pain is highly refractive to existing treatments, and available treatments have serious side effects. Moreover, long-term treatment with analgesics can elicit tolerance, addiction, and sedation. Given these issues, the development of new, safe, and long-lasting pain therapies is of utmost importance.

Previous research has shown that decreases in central inhibition manifested by a decrease in spinal GABA synthesis, and a reduction in the inhibitory postsynaptic potential may underlie a core pathological component of chronic neuropathic pain.[20,21,28,36] Moreover, the spinal intrathecal delivery of GABA and GABA agonists, such as baclofen or muscimol, attenuates neuropathic pain in both rodents[19] and human patients.[18,37]

Currently, clinically approved treatments do not target the underlying causes of neuropathic pain. Given the proposed role of inhibitory neurotransmitters in gating pain[14,21,27,31] at the spinal level and the ability for spinally delivered GABA to alleviate pain, treatments that increase spinal inhibition are likely to be effective analgesics. Other investigators have used foetal progenitor cells[4,5,13] and engineered immortalised cell lines[11] to alleviate chronic pain. However, given their multipotent status or foetal origin, they are unsuitable for clinical use.

We set out to create a safe, therapeutically relevant protocol using matured human induced pluripotent stem cell-derived GABAergic neurons (iGABAergic neurons) and transplant them into the dorsal horn to alleviate pain elicited by spared nerve injury (SNI), a model of persistent neuropathic pain in mice.[10] We found that iGABAergic neurons are able to survive in the SNI-injured spinal cord, and these cells were able to alleviate pain long term (~2 months) with a single treatment. This reflects a viable translational strategy for the alleviation of established neuropathic pain, an area of unmet clinical need.

2. Methods

2.1. Study design

Sample sizes were determined by power calculation that determined based off of our SD; a minimum of 12 mice per group was required for the key study. The results of the main study were used to estimate the required number of mice for the sensory and naïve studies. We used additional mice to perform longer term studies as initial experimental timeframes were limited

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

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by ethical approvals. We did not exclude any outliers; however, any mouse that trapped its paw was not measured for von Frey at that attempt, we performed all calculations including and excluding these mice, and this did not affect the overall result. Overall the key findings of the study were replicated across a total of 6 cohorts of mice. Both the sensory neuron (twice) and naïve mice (3 times) injections were also replicated.

Our a priori hypothesis was that the integration of GABAergic interneurons would alleviate neuropathic pain. We therefore performed baseline behaviour assessments, an SNI and injected iGABAergic neurons into the spinal cord. All experiments involving animals were performed blind (the surgeon performing the laminectomy, SNI, and behaviour was unaware of group assignments). The group assignments were performed before injections by an investigator blind to the behavioural results and health status of all the animals in a pseudorandom manner.

2.2. Animals

All animal experiments were conducted under Animal Ethics Protocol 938 issued by the University of Sydney's Animal Ethics Committee. All experiments comply with the National Health and Medical Research Council of Australia's code for the care and use of animals. The mice used were NOD prkd^{SCID} mice obtained from the Animal Resource Centre, WA. Full details are provided in the supplemental materials (available at http://links.lww.com/PAIN/A897).

2.3. Cells

The ATCC-BXS0116 hiPSC line was used in this study (ATCC, ACS1030).

2.4. Differentiation

Cells were differentiated using existing protocols.[22,35] Full details including adaptations are provided in the supplemental methods (available at http://links.lww.com/PAIN/A897).

2.5. Behaviour assessment

Full details of the von Frey, Acetone, Open-Field, and Gait analysis assays are provided in the supplemental methods (available at http://links.lww.com/PAIN/A897).

2.6. Surgery

Full details of the surgical procedures are provided in the supplemental methods (available at http://links.lww.com/PAIN/A897). Briefly, the SNI was performed according to an existing protocol[10]; full details of the laminectomy are provided in supplemental materials (available at http://links.lww.com/PAIN/A897).

2.7. Flow cytometry

Cells were dissociated with TryPLE (Invitrogen), fixed in 10% formalin (Sigma-Aldrich, HT5011) for 20 minutes and incubated with blocking buffer (PBS with 3% BSA and 0.3% Triton X100). Blocked cells were incubated with primary antibody for 30 minutes in blocking buffer. Cells were washed 3 times with PBS and incubated with Alexa Fluor 594-conjugated secondary antibody for another 30 minutes. After washing with PBS, cells were resuspended in PBS with 3% BSA and analyzed. Mouse IgG isotype control antibodies were used as controls under the same conditions. Unlabelled controls were also used as controls. Full details of antibodies and concentrations are provided in supplemental Table S2 (available at http://links.lww.com/PAIN/A897). Acquisition of 10,000 events was performed using FACSAria (BD Biosciences). Flowjo software was used to analyze raw data.

2.8. Immunofluorescence

Cells were fixed with 10% neutral buffered formalin and stained with antibodies. For mouse immunostaining, mice were perfused, tissue was postfixed, processed by cryosectioning and stained. Full details of the protocol, antibodies used, and their concentrations are provided in Table S1 (available at http://links.lww.com/PAIN/A897).

2.9. Proteomics

The full details of protein extraction, preparation, and proteomic methodology are provided in supplemental methods (available at http://links.lww.com/PAIN/A897).

2.10. RNA sequencing

RNA was extracted by column purification, treated with DNAse on column and sent for RNA sequencing to Novogene. All details of analysis and library preparation are provided in the supplemental methods (available at http://links.lww.com/PAIN/A897) as well as methods for qPCR.

2.11. Calcium imaging

Calcium imaging was adapted from previously published articles.[24,34] Briefly cells were loaded with FURA-2AM and stimulated with glutamate (100 μm with equimolar glycine for NMDA activation). Full details are provided in the supplemental methods (available at http://links.lww.com/PAIN/A897).

2.12. Western blot

A bicinchoninic acid assay was performed according to the manufacturer's instructions, and 10 μg of protein was loaded, electrophoresed, transferred, and stained according to the supplemental methods (available at http://links.lww.com/PAIN/A897).

2.13. ELISA

The ELISA assay was performed on media of 3-day cultured cells from either hiPSC or iGABAergic neurons at 25 days in vitro (DIV).

3. Results

To enhance central inhibitory tone in the mouse spinal cord, we first generated GABAergic interneurons from hiPSC (iGABAergic neurons) using an existing protocol[22] (**Fig. 1A**). Across 28 DIV, we observed efficient differentiation to GAD65/67+ (a marker of GABA synthesis), TUBB3+ (a neuron-specific marker) cells (**Fig. 1B**). We assessed the purity of the 25 DIV cells obtained by flow cytometry, and observed ~95% of cells were GAD65/TUBB3+ (**Fig. 1C**, and Fig. S1, http://links.lww.com/PAIN/A897). We assessed the molecular profile of our transplant materials using

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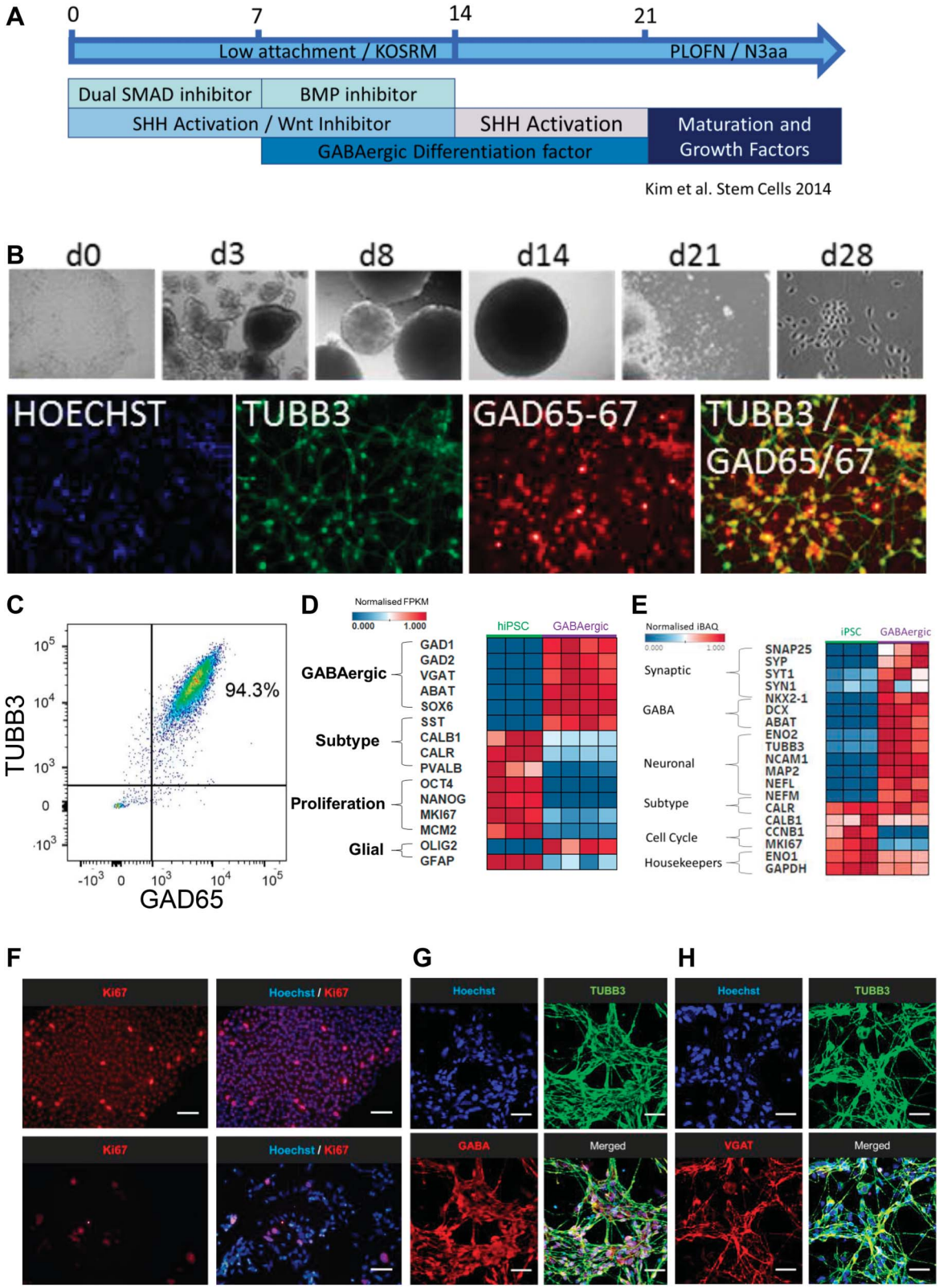

Figure 1. Pure GABAergic neurons can be efficiently derived from human iPSC. (A) Differentiation protocol for the derivation of GABAergic neurons (B) Brightfield images of GABA neurons during differentiation and 28 DIV immunofluorescence images of matured GABAergic neurons expressing typical markers, nuclear DAPI, TUBB3, GAD65/67, and merge channels showing purity. FACS analysis of hiPSC-derived GABA neurons at 25 DIV of differentiation. (C) Dot plot of hiPSC-derived GABA neurons, TUBB3 against GAD65. (D) RNASeq normalised FPKM of GABAergic neurons vs hiPSC; hiPSC (n = 3) and GABAergic neurons (n = 4) from independent differentiations. (E) Proteomics performed on 25 DIV neurons compared with hiPSC (n = 3 per group), Heatmap of normalised iBAQ intensity. (F) Ki67 staining demonstrates few proliferative cells remain at DIV25. (G) Imaging of 25 DIV neurons. Neurons are GABA immunoreactive. (H) Cells express VGAT. DIV, days in vitro.

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RNA sequencing (**Fig. 1D** and Fig. S2, http://links.lww.com/PAIN/A897). We compared our neurons with hiPSC cultures from the same donor and hiPSC-derived sensory neurons (iSensory neurons) to confirm GABAergic specificity. We confirmed the quality of sequencing using hierarchical clustering and noted appropriate clustering and greater clustering between neuronal replicates compared with hiPSC (Fig. S2A, http://links.lww.com/PAIN/A897). Differentiated GABAergic neurons expressed transcripts associated with the synthesis, packaging, and release of GABA as well as markers of GABAergic neuron development (**Fig. 1D**, and Fig. S2B, http://links.lww.com/PAIN/A897). iGABAergic neurons expressed markers of medial ganglion eminence and lateral ganglionic eminence lineage cells (Fig. S3, http://links.lww.com/PAIN/A897). They primarily differentiated into cortical or striatal somatostatin type neurons (Fig. S3, http://links.lww.com/PAIN/A897). We confirmed these observations using qPCR that showed good concordance with our sequencing results (Fig. S2F, http://links.lww.com/PAIN/A897). Since transcriptomic profiles do not always reflect the proteins expressed in a cell,[17] we conducted proteomics using mass spectrometry and identified markers of neurons and GABAergic-specific markers (**Fig. 1E** and Figs. S4A–D, http://links.lww.com/PAIN/A897). We performed Reactome pathway analysis of upregulated proteins, and GABAergic signaling was significantly associated with our iPSC-derived GABAergic cells (Figs. S4C and D, http://links.lww.com/PAIN/A897). Previous studies in this field have used proliferating progenitor cells that may present tumorigenic risks to patients. Therefore, we examined markers of proliferation and pluripotency by transcriptomic and proteomic analysis. We observed sharp declines in Ki67 and were no longer able to observe CCNB1 by proteomics and saw strong decreases in core pluripotency genes by RNA sequencing (Fig. S2D, http://links.lww.com/PAIN/A897). Fluorescence microscopy confirmed a significantly reduced number of Ki67+ cells, and cell proliferation was not observed in concordance with the RNASeq and proteomic results (**Fig. 1F** quantified in Fig. S4E, http://links.lww.com/PAIN/A897). Finally, we confirmed GABA synthesizing machinery (GAD65 positive), the glycinergic neuronal marker (GlyT2 positive) (Fig. S4F, http://links.lww.com/PAIN/A897), and importantly production (**Fig. 1G**), and the GABA transporter required for GABA release (Vesicular GABA/Glycine transporter [VGAT]) (**Fig. 1H**).

Because iGABAergic neurons express GABAergic markers and machinery, we evaluated GABA secretion and confirmed iGABAergic neurons secrete GABA by ELISA (**Fig. 2A**). Moreover, iGABAergic neurons express subunits of kainate, NMDA, and AMPA glutamate receptor subclasses (**Fig. 2B**). To test whether the cells were responsive, we stimulated the cells with glutamate and performed calcium imaging. We observed strong calcium transients and most cells responded to both glutamate and potassium chloride (**Figs. 2C and D**).

To assess the ability of iGABAergic neurons to relieve neuropathic pain, we developed a transplantation protocol to deliver iGABAergic neurons into the dorsal horn of the lumbar enlargement (**Fig. 3A**). NOD scid mice were used to avoid xenograft rejection. Briefly, we performed an SNI, and 1 week later, we transplanted iGABAergic neurons into mice with established neuropathic pain. As expected, SNI caused tactile and cold allodynia in NOD scid mice within 6 days, assessed by a von Frey light touch assay, and acetone response, respectively. From 2 weeks after the transplant, we observed potent analgesia that lasted as long as 2 months (the endpoint for the study) manifesting as a reduction in tactile allodynia and a shift in the stimulus response curves (**Fig. 3B** and Fig. S5A, http://links.lww.com/PAIN/A897). Because GABAergic neurons are also involved in motor circuitry, we assessed motor behaviour using the open-field test. We observed no difference between groups in either the average velocity or the maximal velocity of movement (**Fig. 3D–E**). Furthermore, we assessed the gait of the mice using a modified Basso Mouse Scale. Spared nerve injury had an effect on gait as expected due to the damage to the mixed motor/sensory sciatic nerve. However, we observed no gross change in gait between groups following transplantation, together suggesting no major interference with motor behaviour (**Fig. 3E**). Because of close anatomical proximity, GABAergic neurons implanted into the lumbar dorsal horn have the potential to impact upon autonomic nervous system nuclei located in the spinal cord. However, we did not observe any issues with bladder or bowel movements in our mice. Notably, we did not observe

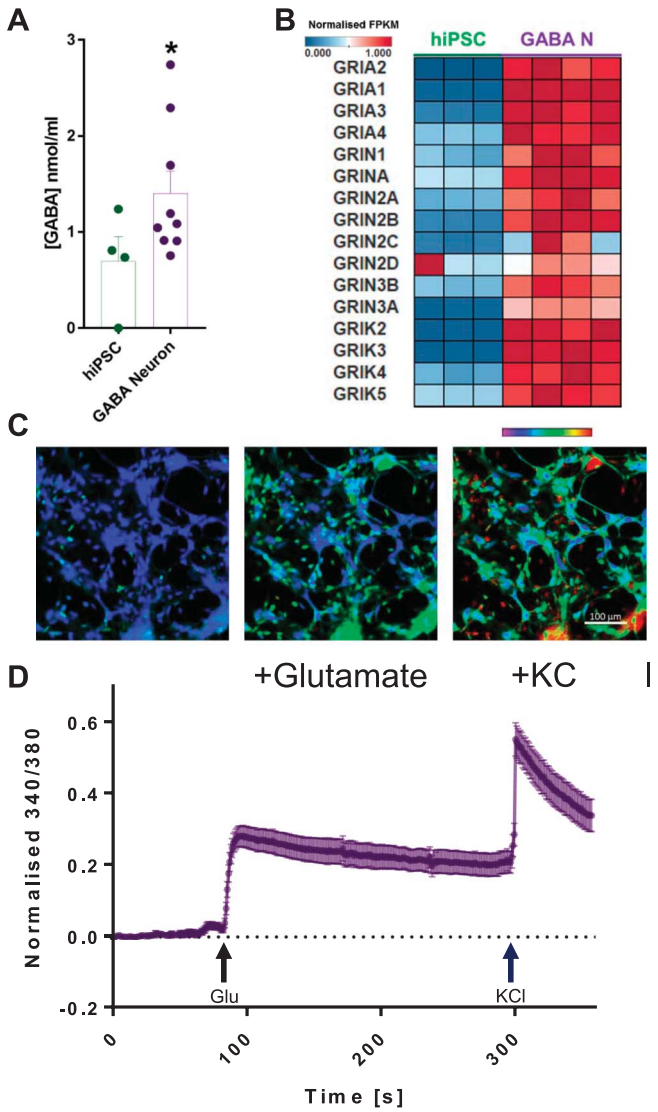

Figure 2. Human iPSC-derived GABAergic neurons are functional in vitro. (A) GABA concentration in media of 25 DIV GABAergic neurons. (B) Normalised FPKM of ionotropic glutamate receptor subunit expression. (C) Representative images of calcium transients following the addition of glutamate and KCl. (D) Quantification of calcium transients exhibited by cells (n = 50). DIV, days in vitro.

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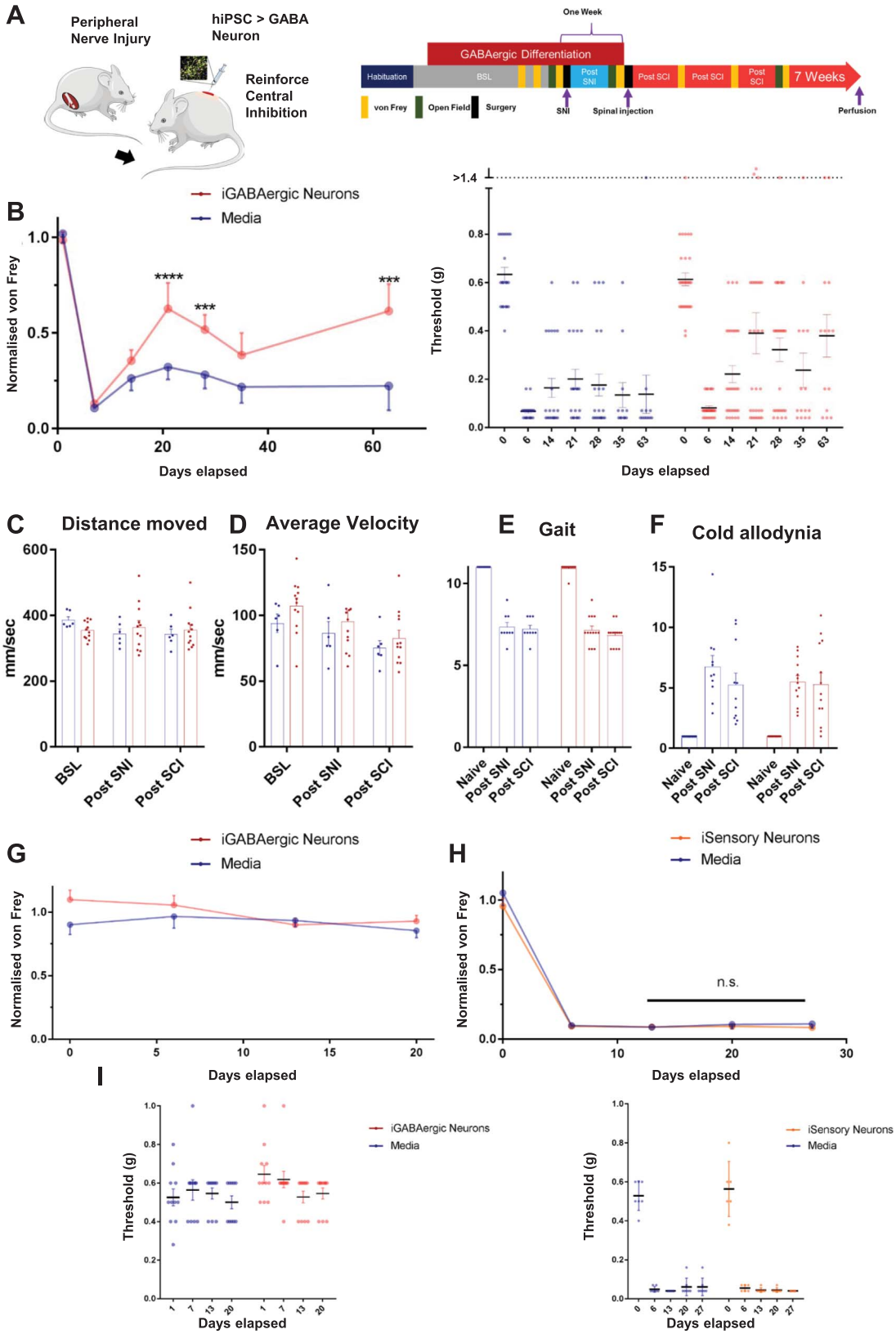

Figure 3. Spinal transplantation of human iPSC-derived GABAergic neurons alleviates tactile allodynia. (A) Schematic of strategy of laminectomy-based spinal insertion of GABAergic neurons into L1 following spared nerve injury (peripheral neuropathic pain model) and timeline of in vivo experimentation. (B) Normalised von Frey thresholds of injured mice after nerve injury and spinal transplantation of GABAergic neurons (n = 21 vehicle, 29 GABAergic neurons). (C) Stimulus response curves relative to nerve-injured mice at 2 months (n = 12 vehicle and n = 16 GABAergic neurons) (D and E), open-field motor analysis of mice following experimentation using open-field paradigm, and (F) ipsilateral modified Basso Mouse gait analysis scale scores. (G) Acetone cold allodynia responses. (H) Normalised von Frey thresholds following experiments in naïve mice (n = 11). (I) Normalised von Frey thresholds after nerve injury and injection of iSensory neurons (n = 7). Multiple comparisons performed using repeated-measures ANOVA with Bonferroni correction for (D–I). Two-way ANOVA for B with Bonferroni correction. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$. Supplemental Figure 5 displays full data distributions (available at http://links.lww.com/PAIN/A897). ANOVA, analysis of variance.

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analgesia in the acetone test suggesting either a lack of efficacy of the cells to alleviate this behaviour and/or a different spinal aetiology (**Fig. 3F**). To further confirm the effects were unrelated to motor behaviour, we transplanted iGABAergic neurons into naïve mice; we found no change in both stimulus response and withdrawal threshold suggesting a nerve injury is required for the reduction in tactile sensitivity (**Fig. 3G** and Fig. S5B, http://links.lww.com/PAIN/A897). These results also suggested that transplants do not simply increase the tonic threshold at which any stimulus is perceived (Fig. S5B, http://links.lww.com/PAIN/A897). Finally, we confirmed the specificity of our system since transplanting hiPSC-derived sensory neurons (iSensory neurons) into the dorsal horn after nerve injury were unable to elicit analgesia (**Fig. 3H** and Fig. S5C, http://links.lww.com/PAIN/A897). Together, enhancing inhibition using mature iGABAergic neuron transplants can promote long-term analgesia from peripherally induced neuropathic pain.

To assess the potential for neurons to engraft in the spinal cord, we assessed the cells' survival, integration, and potential for synaptic integration (Fig. S6, http://links.lww.com/PAIN/A897). Three weeks after transplantation, we identified hiPSC-derived GABAergic neurons in the dorsal horn laminae I (CGRP+) and II (IB4+) as well as other dorsal laminae through radial migration (Fig. S6, http://links.lww.com/PAIN/A897). Initially, iGABAergic neurons did not migrate to ventral or contralateral spinal cord (Fig. S6, http://links.lww.com/PAIN/A897), retained neuronal, and GABAergic marker expression and showed early evidence of synaptic integration potential (Fig. S6, http://links.lww.com/PAIN/A897). However, after 10 weeks, we observed substantial migration. Transplanted cells retained their neuronal identity as assessed by their coexpression of human NCAM1, MAP2, TUBB3, and expressed NeuN suggesting terminal differentiation (**Figs. 4C–E**). iGABAergic neurons were immunoreactive to GABA and retained VGAT, GAD65/67, and synapsin expression indicating transplanted iGABA neurons retain the ability to synthetise, package, and release GABA (**Figs. 4F–H**). In addition, we found mouse presynaptic densities (marked by a mouse-specific antibody targeting Bassoon and colocalising with presynaptic protein RIM2) were in direct apposition to iGABAergic neurons (marked by a human specific cytoplasmic antibody, HuCytoplasm), suggesting the potential to form synapses between transplant and recipient tissue (**Fig. 4I**). iGABAergic neurons expressed critical presynaptic proteins, liprin, and were immunoreactive to a pan-voltage-gated calcium channel antibody (**Figs. 4K and L**). Finally, we observed the apposition of the inhibitory postsynaptic marker gephyrin with human synapsin suggesting the presence of inhibitory synapses (**Figs. 4M and N**). Of note, the transplanted inhibitory neurons were predominately somatostatin or parvalbumin subclasses (**Figs. 4O and P**). We did not observe any morphological evidence of teratoma or other related abnormalities, which suggests the safety of our procedure, and identified no evidence that iGABAergic neurons are proliferating (did not express active cell cycle protein Ki67) (**Figs. 4R and S**).

4. Discussion

Together we provide the first evidence that iPSC-derived GABAergic neurons can be used for the treatment of established neuropathic pain. We were able to efficiently derive GABAergic neurons from hiPSC, these cells survived when transplanted into the mouse spinal cord, they integrated into the cord and formed potential synapses. This work extends upon previous observations[4,13] by moving away from the use of multipotent mouse precursor cells and toward the use of therapeutically relevant human terminally differentiated materials. These transplants are safe, seem to be pain-specific, and do not have overt effects on motor function.

Our data are in line with multiple recent studies that have transplanted primary mouse GABAergic cortical interneuron precursors obtained from the foetal medial ganglionic eminence (MGE) to treat epilepsy,[1,2] neuropathic pain, chemotherapy-induced peripheral neuropathy, and pathological itch.[4,5] More recently, MGE progenitor-like transplants generated from human embryonic stem cells have also been shown to provide long-lasting relief in mice with pain or dysfunction resulting from spinal cord injury.[13] These studies established a proof of principle for neuronal cell therapies to treat pain. However, MGE progenitors are multipotent and thus can differentiate into other cell types. Moreover, human primary MGE progenitor transplants are not possible, and transplantation of embryonic stem cell-derived MGE progenitor pools would require long-term immune suppression and therefore elevation of the risk of cancer.

Consistent with previous studies using GABAergic neural precursors, we found that increasing GABAergic inhibition through transplantation does not increase baseline tactile thresholds.[4,13] Similarly, increasing inhibition by activation of KCC2 and the subsequent restoration of spinal chloride balance also typically has no effect on baseline thresholds.[9,15] However, the pharmacogenetic activation of endogenous GABAergic interneurons trigger an increase in baseline thresholds.[29] This apparent divergence between GABAergic inhibitory potential and functional effects could suggest that evoked GABA release rather than tonic inhibition plays a central role in the gating of tactile stimuli in the spinal cord. Alternatively, distinct levels of GABAergic inhibition may regulate different pathways with greater tonic inhibition required to inhibit tactile rather than nociceptive pathways. Further detailed studies will be essential to elucidate the complexities of this spinal processing.

Although iGABAergic neurons show considerable efficacy against neuropathic pain, we have only thus far assessed effects in the SNI model; the use of other models of neuropathic pain will be essential to ensure translation. Necessarily, we used an immune-deficient mouse for the transplantations, but neuropathic pain reportedly has a strong immunological component.[3,6,26,32] Importantly, other groups have demonstrated efficient integration of mouse-derived cells in immunocompetent mice.[4,5,12,23] Furthermore, although we did inject into mice with established neuropathic pain, many patients suffer from pain for years, and further work is required to assess iGABAergic neuronal transplant efficacy in groups with entrenched chronic pain. Finally, to assess the integrative potential of human-derived material, more detailed studies will be required to assess both the function and structure of the putative synapses by electron microscopy and electrophysiology.

Differentiated iGABAergic neurons have far greater clinical relevance than previous approaches because they can be used for autologous transplants.[25,33] For further preclinical development, more detailed safety and efficacy studies are required in larger animal models and other rodent models. More detailed work will be necessary to establish site safety along the spinal cord given the anatomical heterogeneity.[7,30] Furthermore, the development of high-quality good manufacturing practice material for safe transplantation will be essential. Existing analgesics have a limited therapeutic time-window, and some have serious adverse effects that are exacerbated by repeated dosing. This therapy represents the possibility of a single procedure that

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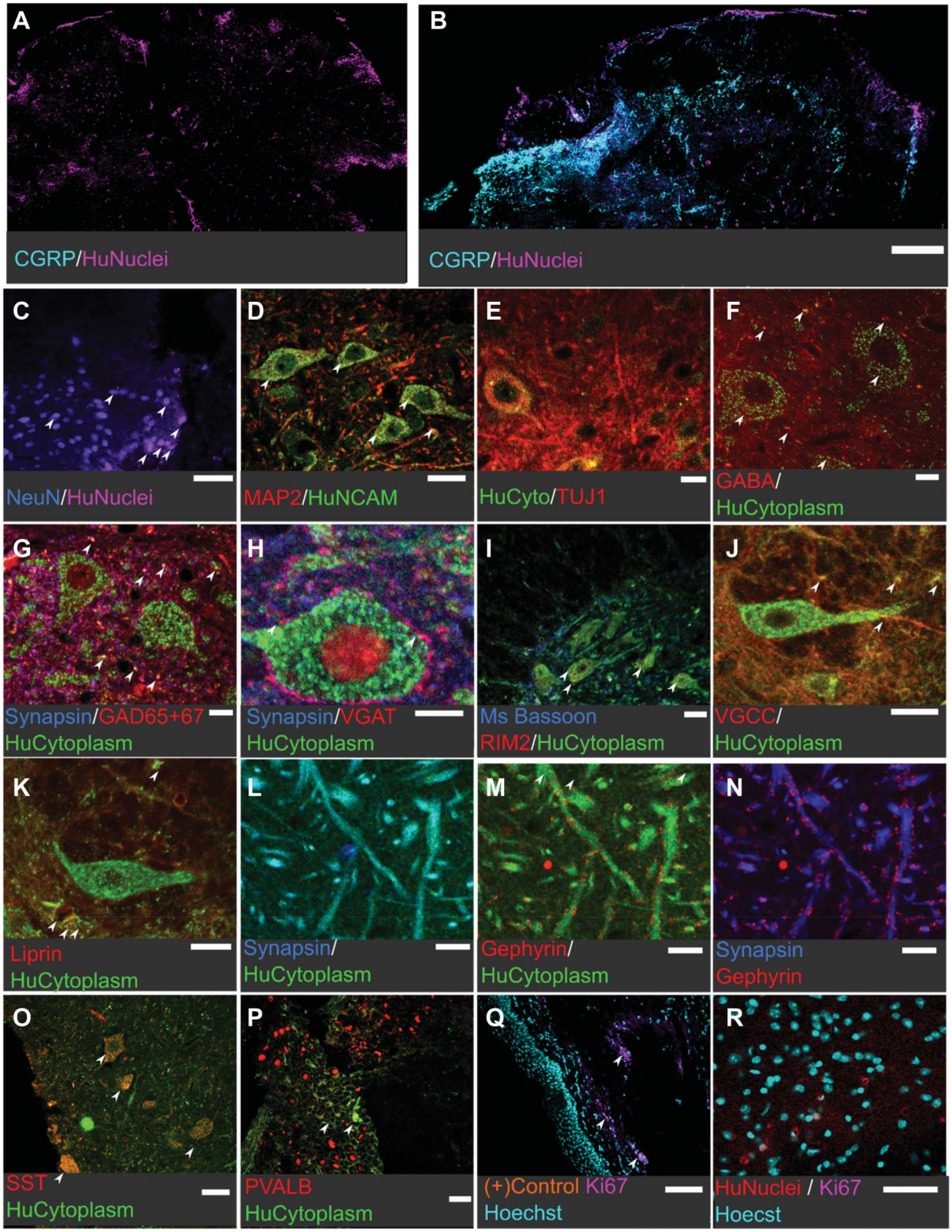

Figure 4. Human iPSC-derived GABAergic transplants survive and can integrate with endogenous circuitry after 10 weeks. (A) Cells migrate extensively. (B) Cells locate within the dorsal horn. (C) Human nuclei colocalise with NeuN. (D) Human neuronal adhesion molecule colocalises with MAP2. (E) Human cytoplasm colocalises with TUBB3. (F) GABA is found in human cytoplasm-positive cells. (G) GAD65/67 is found at putative human synapses. (H) Human cell located synaptic vesicles are positive for VGAT. (I) Human cells are apposed to Mouse Bassoon, RIM2. (J) Voltage gated calcium channels (VGCC) are found in human cells. (K) Liprin is found at putative human synapses. (L–N) Human cytoplasm located synapsin is located apposed to Gephyrin suggesting postsynaptic development. (O) SST is found in human cells. (P) Parvalbumin is found in human cells. (Q–R) Ki67 staining in positive control mouse skin and not in human nuclei in spinal cord. Scale bars: 30 μm = (C), (I and Q), 20 μm = (D), 15 μm = (E–H) 10 μm = (G–L).

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provides longer lasting analgesia and effectively alleviates neuropathic pain without side effects.

Neuropathic pain is difficult to treat and has serious impacts on daily function, lifestyle, and mood. Peripheral nerve injuries cause a decrease in GABAergic inhibition at the level of the spinal cord. Reinforcing central inhibition by transplanting safe, terminally differentiated GABAergic neurons, is able to alleviate pain and should be considered as a new strategy to manage chronic pain.

Conflict of interest statement

J. Manion, L. Caron, and G.G. Neely are inventors on a PCT patent application for the technology described in this article. The remaining authors have no conflicts of interest to declare.

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Transcriptomic data are available from GEO: GSE115565.
Proteomics data are available at ProteomeXchange PXD009926
All other data are available from the corresponding authors.

Appendix A. Supplemental digital content

Supplemental digital content associated with this article can be found online at http://links.lww.com/PAIN/A897.

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